

IRRIGATION CIRCUIT FOR THE CONTROL OF SOIL MOISTURE

DOCUMENTATION

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1. Table of Contents.

1. Table of Contents.....	1
2. Specifications.....	2
2.1 Project overview.....	2
2.2 Functionality of the circuit.....	3
2.3 Design requirements.....	3
3. Block Diagram.....	3
4. Electrical Schematic.....	4
4.1 Sensor resistance stage & Conversion.....	5
4.1.1 Circuit snippet of the stage.....	5
4.1.2 Theoretical approach.....	5
4.1.3 Mathematical approach for power consumption.....	7
4.1.4 Mathematical proof.....	7
4.1.5 Simulation of the sensor.....	10
4.2 Comparator stage (Schmitt Trigger).....	11
4.2.1 Circuit snippet of the stage.....	12
4.2.2 Theoretical overview.....	12
4.2.3 Standardization of resistors.....	13
4.2.4 Mathematical approach, simulations.....	14
4.3 Buffer.....	18
4.3.1 Circuit snippet of the stage.....	18
4.3.2 Theory approach.....	18
4.3.3 Simulation.....	19
4.4 Power electronics stage.....	19
4.4.1 Power connections snippet (relay, LED, BJT).....	20
4.4.2 Theory approach, and calculation for LED.....	20
4.4.3 Simulations for LED.....	21
4.4.4 Theory approach, and calculation for relay.....	22
4.4.5 Simulation for relay.....	23
5. Monte Carlo Statistical Analysis.....	23
6. Worst Case Analysis.....	26

7. Design Enhancements.....	28
7.1 Initial circuit.	29
7.1.1 Modifications.....	29
7.2 Final circuit.....	30
8. Bill of Materials.	31
9. Bibliography.	35

2. Specifications.

2.1 Project overview.

The following document contains the pursuit of designing a tailored soil irrigation system dedicated to the automated maintenance of plants within an enclosed range percentage. It depends on the bare humidity of the ground, and it is mainly used in agriculture for the detection of surface moisture fluctuation levels. Implicitly, the sensor technology must be able to consistently bypass irrigation at a predetermined threshold, allowing the soil to be watered strictly when it is required. This allows for complete autonomy from human intervention.

2.2 Functionality of the circuit.

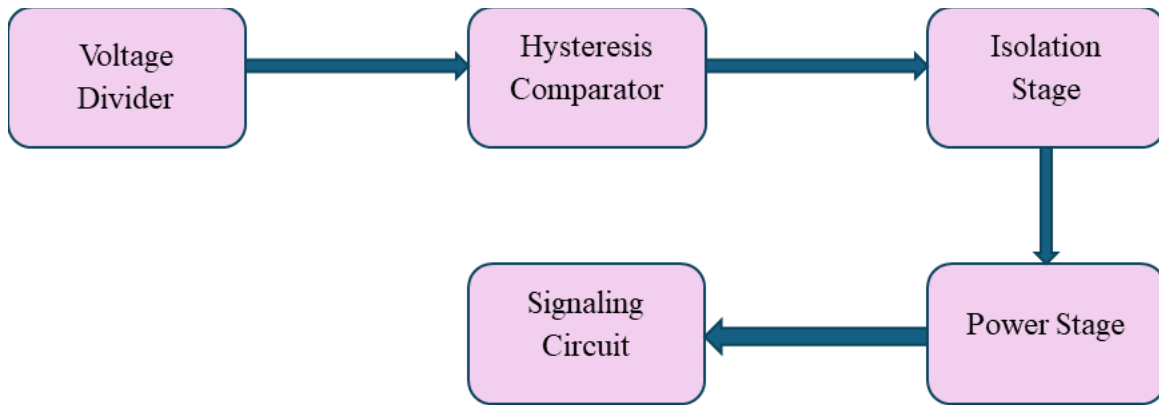
Given that the moisture sensor measures humidity linearly within a 15%–50% range, the system is set to strictly preserve soil moisture within the 15%–30% interval. To achieve this, the linear variation of the sensor's resistance (950 k Ω – 55 k Ω) is converted into a voltage signal within the range of $[0 \div (V_{cc}-2V)]$, relative to a 13V supply voltage.

Furthermore, the soil moisture is maintained within the target domain using a comparator and an electromagnetic relay that controls the water pump, which is modeled for simulation purposes as a resistive load. Finally, the operational status of the pump is indicated by a blue LED, which illuminates during active irrigation and ceases once the desired moisture level is reached.

2. 3 Design requirements.

<i>Humidity level to maintain [%]</i>	<i>Maximum humidity range [%]</i>	<i>Sensor resistance [kΩ]</i>	<i>VCC [V]</i>	<i>LED Color</i>
15 - 30	15 - 50	950-55	13	BLUE

3. Block Diagram.



4. Electrical Schematic.

4.1 Sensor resistance stage & Conversion.

4.1.1 Circuit snippet of the stage.

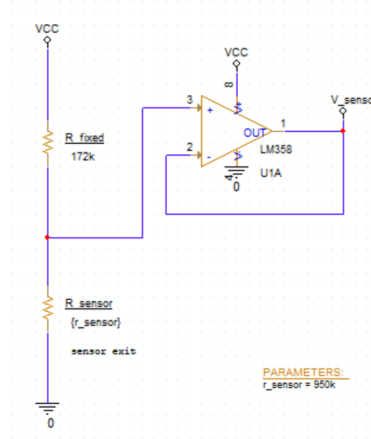


Figure 1. Sensor resistance stage.

4.1.2 Theoretical approach.

To initiate the project, I began by polarizing the sensor with a 13V DC supply so that the sensor has the energy required (generated by the electric field) to translate the real-life moisture into an electrical signal I can measure. Furthermore, to convert the sensor's variable resistance into a readable voltage, I implemented a voltage divider network.

$$i = \frac{v_I}{R_1 + R_2} \quad (1)$$

$$v_O = i \times R_2 = \frac{R_2}{R_1 + R_2} \times v_I \quad (1.2)$$

Formula 1. General Voltage Divider Deduction

The sensor's resistance is set as a global parameter {r_sensor} so that we can observe how it varies when simulating. This varies between 950 kΩ (at 15% humidity) and 55 kΩ (at 50% humidity).

In my case, the V_{sensor} at the divider node is determined similarly to the formula (1). Therefore, as my soil moisture increases $\{r_{\text{sensor}}\}$ drops from 950 k Ω down to 55 k Ω , which relates to the formula below:

$$V_{\text{sensor}} = V_{CC} \times \frac{R_{\text{sensor}}}{R_{\text{sensor}} + R_{\text{fixed}}} \quad (2)$$

Formula 2. Voltage Divider Formula

The main issue according to the *Power Electronics Handbook* edited by Rashid¹ is the loading effect. It occurs when the input impedance of the next stage is low enough to absorb the current from the sensor, which leads to an undesired voltage drop to take place. The comparator (next stage) disturbs the calculated voltage which will undoubtedly compromise the simulations.

To mitigate this, I implemented an impedance isolation buffer before connecting it to the comparator. Consequently, this connect allows for the signal to be picked over and repeated at the output with the theoretical value. This can be possible because of the op-amp 's high impedance which prevents any current draw from the voltage divider.

For my OP-AMPS, I wanted to use a component that would be strictly tailored to high efficiency. This idea came to mind when I considered the cost of the overall circuit. Initially, I chose an L272 because, based on the datasheet, it had features that would have been a strong contender for my project with a range of [4V;28V] and 1A of output current. Therefore, it could have driven the relay straight from the output pin without the need for a transistor.

The problem arose when I took into account the cost as well as the power consumption. Firstly, I looked for a similar transistor which I could compare to my L272, and I found the LM358.

The LM358 ranges around the same price and voltage thresholds, yet it differs at the supply current. This caught my attention, and I underwent a comparison process.

OP AMP	PRICE	SUPPLY CURRENT
LM358	5 RON	0.7 mA
L272	8 RON	10 mA

¹ M. H. Rashid, Ed., *Power Electronics Handbook*. San Diego, CA: Academic Press, 2001.

Following the take on power loss by Rashid² in his book, I calculated the power dissipation P_Q for each op-amp.

$$P_Q = V_{cc} \times I_Q \quad (3)$$

Formula 3. Quiescent Power Dissipation

4.1.3 Mathematical approach for power consumption.

$$\text{LM358} \rightarrow I_Q = 0.7\text{mA}$$

$$P_Q = V_{cc} \times I_Q \rightarrow P_Q = 13\text{V} \times 0.7\text{mA} \rightarrow$$

$$P_Q = 9.1\text{mW}$$

$$\text{L272} \rightarrow I_Q = 10\text{mA}$$

$$P_Q = V_{cc} \times I_Q \rightarrow P_Q = 13\text{V} \times 10\text{mA} \rightarrow$$

$$P_Q = 130\text{mW}$$

We can approximate that the LM358 is 14 times more efficient than the L272. Consequently, as stated by Rashid, “we minimize power consumption to extend battery life”³, and so I chose as my definitive answer the LM358 operational amplifier. This is paramount, hence, not only do we minimize power consumption but also the cost. Having taken into account this factor, my battery life will have longer longevity, and therefore, my circuit will be cheaper.

4.1.4 Mathematical proof.

We will use formula (1.2) to determine the value of the resistance (R_{fixed}) for my voltage divider.

$$V_{out} = V_{cc} \times \frac{R_{sensor}}{R_{sensor} + R_{fixed}} \rightarrow$$

² M. H. Rashid, Ed., *Power Electronics Handbook*. San Diego, CA: Academic Press, 2001.

³“Applications of DC-DC Converters,” in *Power Electronics Handbook*, M. H. Rashid, Ed. San Diego, CA: Academic Press, 2001, p. 233.

$$\rightarrow 11 = 13 \times \frac{950}{950 + R_{fixed}} \rightarrow \frac{11}{13} = \frac{950}{950 + R_{fixed}}$$

$$\rightarrow 0.846 (950 + R_{fixed}) = 950 \rightarrow$$

$$\rightarrow 803.84 + 0.846 \times R_{fixed} = 950 \rightarrow$$

$$\rightarrow R_{fixed} \cong 172 \text{ k}\Omega$$

This tailored value will help me get the voltage I need at the output of the first stage, respectively, 11V.

However, having tackled that, it is of utmost importance to now determine the resistance at which the irrigation pump turns off. In my project requirements, it is explicit that it turns off at 30%. Because the sensor behaves linearly, I decided to calculate the value by finding the slope of the line. For this, I used the help of the webpage <https://interpolationcalculator.com/>.

Albeit, this will only be used to prove the calculations I have made prior.

$$y = mx + b \quad (4)$$

Formula 4. Slope Formula

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{55 - 950}{50 - 15} = -25.6\%$$

Therefore, for each 1% humidity we drop the resistance by 25.6%.

The difference, from my starting point of irrigation (15% humidity) to my ending (30% humidity):

$$\Delta \text{humidity} = 30 - 15 = 15 \%$$

$$R_{30} = 950 - (15 \times 25.6) \rightarrow$$

$$\rightarrow R_{30} \cong 566 \text{ k}\Omega$$

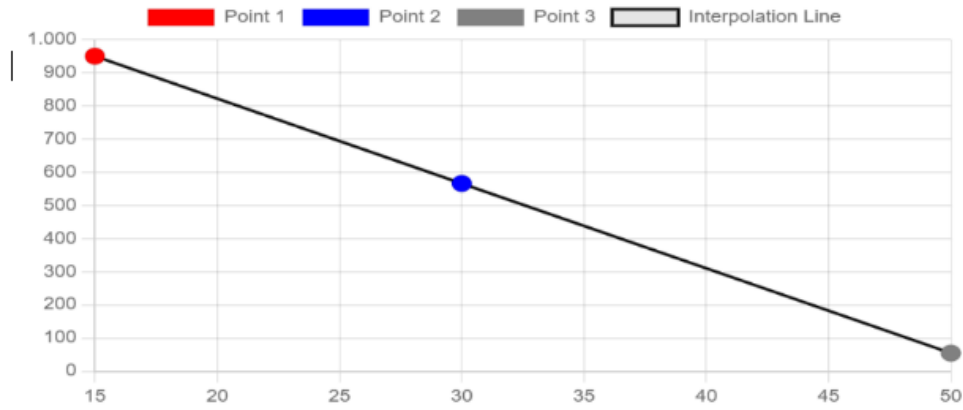


Figure 2. Visual Linear Interpolation

However, there must be determined the voltage range for the extreme values for my sensor [55kΩ – 950kΩ]. For this I will use the following equation:

$$V_{range} = \frac{R_{sensor}}{R_{sensor} + R_{fixed}} \times V_{cc} \quad (5)$$

Formula 5. Voltage Divider

$$15\% \rightarrow R_{sensormin} = 55k\Omega$$

$$V_{rangemin} = \frac{R_{sensormin}}{R_{sensormin} + R_{fixed}} \times V_{cc} \rightarrow$$

$$V_{rangemin} = \frac{55}{55 + 172} \times 13 = 3.14V$$

$$950\% \rightarrow R_{sensormax} = 950k\Omega$$

$$V_{rangemax} = \frac{R_{sensormax}}{R_{sensormax} + R_{fixed}} \times V_{cc} \rightarrow$$

$$V_{range\ max} = \frac{950}{950 + 172} \times 13 = 11V$$

For my voltages at 15%, and 30%, I applied Millman's Theorem:

$$V_{in} = \frac{\frac{V_{cc}}{R_{fixed}} + \frac{GND(0)}{R_{sensor}}}{\frac{1}{R_{fixed}} + \frac{1}{R_{sensor}}} \quad (6)$$

Formula 6. Millman's Theorem

$$15\% \rightarrow R_{\text{sensor}} = 950k\Omega$$

$$V_{th} = \frac{\frac{13}{172}}{\frac{1}{172} + \frac{1}{950}} = 11.033V$$

$$30\% \rightarrow R_{30} = 566k\Omega$$

$$V_{th30} = \frac{\frac{13}{172}}{\frac{1}{172} + \frac{1}{566}} = 9.988V$$

Those two voltages are of utmost importance hence they will confirm the functionality of the sensor.

4.1.5 Simulation of the sensor.

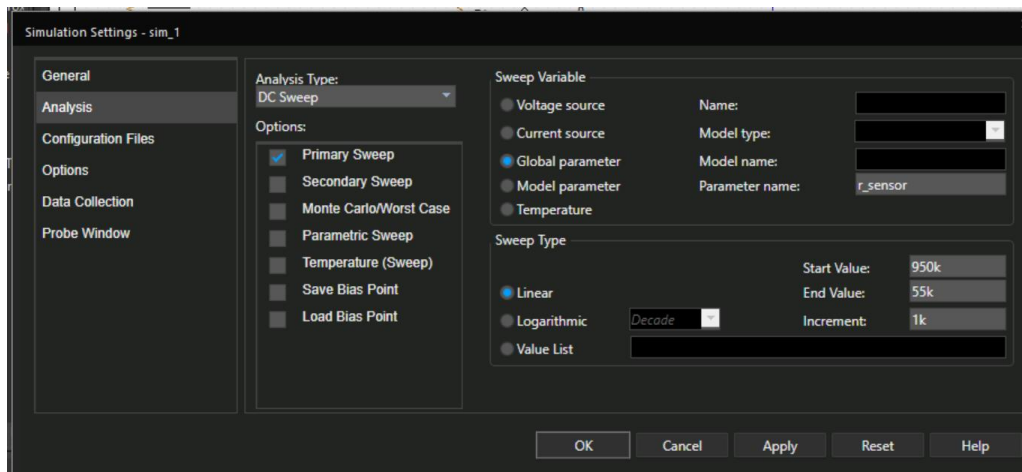


Figure 3. DC sweep parameters

I realized a DC Sweep in order to be able to observe how the voltage varies with respect to the value of the sensor. I selected a “global parameter”, hence my voltage varies based on some parameters. Because I want to see more points in my waveform, I selected a small increment, in this way the representation will be more accurate.

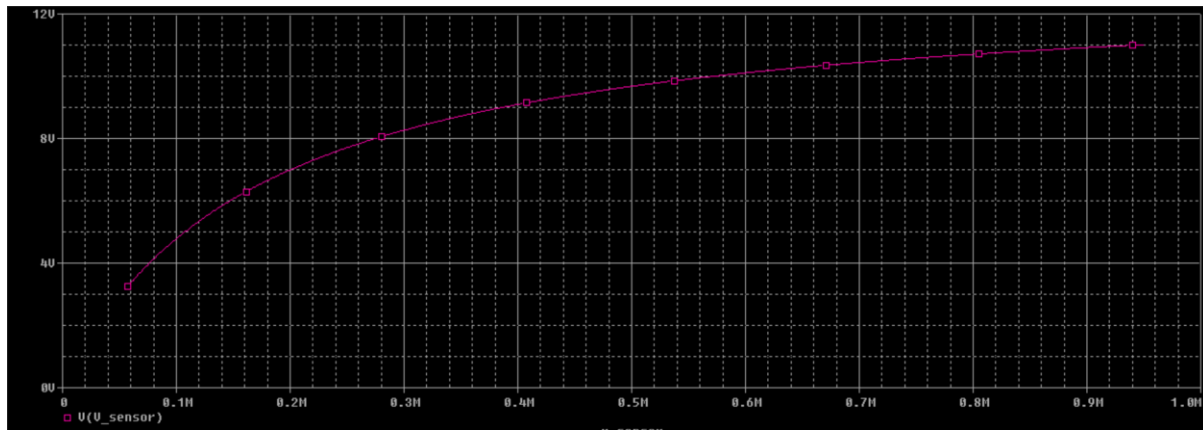


Figure 4. Voltage through the sensor.

We can observe on the graphic how my sensor rises from $\cong 3.1V$ ($55k\Omega$) and ends at $\cong 11V$ ($950k\Omega$).

4.2 Comparator stage (Schmitt Trigger).

4.2.1 Circuit snippet of the stage.

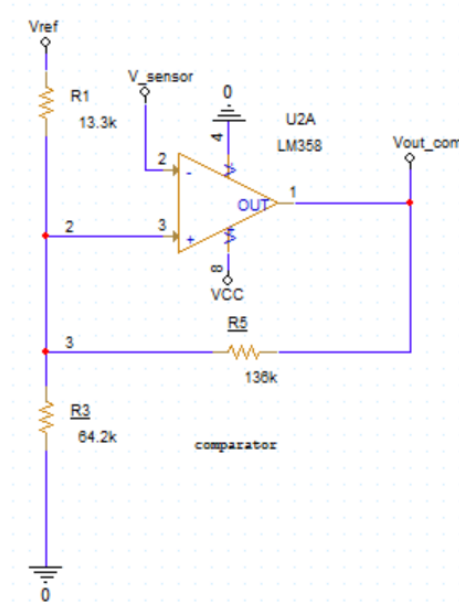


Figure 5. Comparator stage.

4.2.2 Theoretical overview.

In this stage I prioritized simplicity, thus I applied a single non-symmetrical Schmitt trigger connection to keep the circuit as straight-forward as possible. This connection uses an inverting single powered Schmitt trigger where $V_{ref} = V_{cc}$. We use Schmitt Trigger to remove noise from the signal by using hysteresis and therefore reproduce it, as it is one of its main applications.

In section 3.1.4 I calculated the threshold voltages at 15% and 30%, which will now help me calculate the hysteresis window. This window is the variation where the part switches on the rising edge and vice versa.

$$\Delta V_{hys} = V_{th} - V_{th30} \quad (7)$$

Formula 7. Hysteresis window.

Unlike a plain comparator, the Schmitt trigger introduces a hysteresis between an upper and lower threshold. The reference voltage V_{ref} applied to the inverting input is derived from the main supply V_{cc} using a standardized voltage divider.

4.2.3 Standardization of resistors.

In order to have a better chance of producing this circuit in real life, we must use components that are on the market, and not only theoretical values. To standardize the values, I used the table below:

Standard resistor values and Tolerances												
20%	1.00	1.50	2.20	3.30	4.70	6.80						
10%	1.00	1.20	1.50	1.80	2.20	2.70	3.30	3.90	4.70	5.60	6.80	8.20
5%	1.00	1.10	1.20	1.30	1.50	1.60	1.80	2.00	2.20	2.40	2.70	3.00
	3.60	3.90	4.30	4.70	5.10	5.60	6.20	6.80	7.50	8.20	9.10	
2%	1.00	1.05	1.10	1.15	1.21	1.27	1.33	1.40	1.47	1.54	1.62	1.69
	1.87	1.96	2.05	2.15	2.26	2.37	2.49	2.61	2.74	2.87	3.01	3.16
	3.48	3.65	3.83	4.02	4.22	4.42	4.64	4.87	5.11	5.36	5.62	5.90
	6.49	6.81	7.15	7.50	7.87	8.25	8.66	9.09	9.53			
1%	1.00	1.02	1.05	1.07	1.10	1.13	1.15	1.18	1.21	1.24	1.27	1.30
	1.37	1.40	1.43	1.47	1.50	1.54	1.58	1.62	1.65	1.69	1.74	1.78
	1.87	1.91	1.96	2.00	2.05	2.10	2.15	2.21	2.26	2.32	2.37	2.43
	2.55	2.61	2.67	2.74	2.80	2.87	2.94	3.01	3.09	3.16	3.24	3.32
	3.48	3.57	3.65	3.74	3.83	3.92	4.02	4.12	4.22	4.32	4.42	4.53
	4.75	4.87	4.99	5.11	5.23	5.36	5.49	5.62	5.76	5.90	6.04	6.19
	6.49	6.65	6.81	6.98	7.15	7.32	7.50	7.68	7.87	8.06	8.25	8.45
	8.66	8.87	9.09	9.31	9.53	9.76						
0.5%	1.00	1.01	1.02	1.04	1.05	1.06	1.07	1.09	1.10	1.11	1.13	1.14
0.25%	1.17	1.18	1.20	1.21	1.23	1.24	1.26	1.27	1.29	1.30	1.32	1.33
0.1%	1.37	1.38	1.40	1.42	1.43	1.45	1.47	1.49	1.50	1.52	1.54	1.56
	1.60	1.62	1.64	1.65	1.67	1.69	1.72	1.74	1.76	1.78	1.80	1.82
	1.87	1.89	1.91	1.93	1.96	1.98	2.00	2.03	2.05	2.08	2.10	2.13
	2.18	2.21	2.23	2.26	2.29	2.32	2.34	2.37	2.40	2.43	2.46	2.49
	2.52	2.58	2.61	2.64	2.67	2.71	2.74	2.77	2.80	2.84	2.87	2.91
	2.94	2.98	3.01	3.05	3.09	3.12	3.16	3.20	3.24	3.28	3.32	3.36
	3.40	3.44	3.48	3.52	3.57	3.61	3.65	3.70	3.74	3.79	3.83	3.88
	3.92	3.97	4.02	4.07	4.12	4.17	4.22	4.27	4.32	4.37	4.42	4.48
	4.53	4.59	4.64	4.70	4.75	4.81	4.87	4.93	4.99	5.05	5.11	5.17
	5.23	5.30	5.36	5.42	5.49	5.56	5.62	5.69	5.76	5.83	5.90	5.97
	6.04	6.12	6.19	6.26	6.34	6.42	6.49	6.57	6.65	6.73	6.81	6.90
	6.98	7.06	7.15	7.23	7.32	7.41	7.50	7.59	7.68	7.77	7.87	7.96
	8.06	8.16	8.25	8.35	8.45	8.56	8.66	8.76	8.87	8.98	9.09	9.19
	9.31	9.42	9.53	9.65	9.76	9.88						

Figure 6. Values for any tolerance for resistors.

We will see later how, based on the tolerance I will use for my statistical analysis, the resistor will be chosen from the values for each percentage. For this moment the resistors will be standardized from the E24 table which corresponds to 5% tolerance:

Nominal values of resistances							
0.01 Ω	0.1 Ω	1 Ω	10 Ω	100 Ω	1 k Ω	10 k Ω	100 k Ω
0.011 Ω	0.11 Ω	1.1 Ω	11 Ω	110 Ω	1.1 k Ω	11 k Ω	
0.012 Ω	0.12 Ω	1.2 Ω	12 Ω	120 Ω	1.2 k Ω	12 k Ω	
0.013 Ω	0.13 Ω	1.3 Ω	13 Ω	130 Ω	1.3 k Ω	13 k Ω	
0.015 Ω	0.15 Ω	1.5 Ω	15 Ω	150 Ω	1.5 k Ω	15 k Ω	
0.016 Ω	0.16 Ω	1.6 Ω	16 Ω	160 Ω	1.6 k Ω	16 k Ω	
0.018 Ω	0.18 Ω	1.8 Ω	18 Ω	180 Ω	1.8 k Ω	18 k Ω	
0.02 Ω	0.2 Ω	2.0 Ω	20 Ω	200 Ω	2.0 k Ω	20 k Ω	
0.022 Ω	0.22 Ω	2.2 Ω	22 Ω	220 Ω	2.2 k Ω	22 k Ω	
0.024 Ω	0.24 Ω	2.4 Ω	24 Ω	240 Ω	2.4 k Ω	24 k Ω	
0.027 Ω	0.27 Ω	2.7 Ω	27 Ω	270 Ω	2.7 k Ω	27 k Ω	
0.03 Ω	0.3 Ω	3.0 Ω	30 Ω	300 Ω	3.0 k Ω	30 k Ω	
0.033 Ω	0.33 Ω	3.3 Ω	33 Ω	330 Ω	3.3 k Ω	33 k Ω	
0.036 Ω	0.36 Ω	3.6 Ω	36 Ω	360 Ω	3.6 k Ω	36 k Ω	
0.039 Ω	0.39 Ω	3.9 Ω	39 Ω	390 Ω	3.9 k Ω	39 k Ω	
0.043 Ω	0.43 Ω	4.3 Ω	43 Ω	430 Ω	4.3 k Ω	43 k Ω	
0.047 Ω	0.47 Ω	4.7 Ω	47 Ω	470 Ω	4.7 k Ω	47 k Ω	
0.051 Ω	0.51 Ω	5.1 Ω	51 Ω	510 Ω	5.1 k Ω	51 k Ω	
0.056 Ω	0.56 Ω	5.6 Ω	56 Ω	560 Ω	5.6 k Ω	56 k Ω	
0.062 Ω	0.62 Ω	6.2 Ω	62 Ω	620 Ω	6.2 k Ω	62 k Ω	
0.068 Ω	0.68 Ω	6.8 Ω	68 Ω	680 Ω	6.8 k Ω	68 k Ω	
0.075 Ω	0.75 Ω	7.5 Ω	75 Ω	750 Ω	7.5 k Ω	75 k Ω	
0.082 Ω	0.82 Ω	8.2 Ω	82 Ω	820 Ω	8.2 k Ω	82 k Ω	
0.091 Ω	0.91 Ω	9.1 Ω	91 Ω	910 Ω	9.1 k Ω	91 k Ω	

Figure 7. E24 value list.

Furthermore, it must be mentioned that the standardization does not change the functionality or simulation of the circuit. We do this with the purpose of implementing the circuit in a real-life situation.

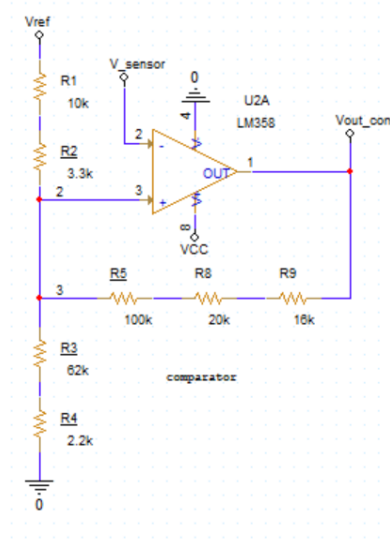


Figure 8. Comparator stage with standardized resistor values.

4.2.4 Mathematical approach, simulations.

We start by using formula (7) to calculate our hysteresis window:

$$\begin{aligned}\Delta V_{hys} &= V_{th} - V_{th30} = 11.03V - 9.988V = \\ &= 1.035V\end{aligned}$$

This result is the distance between the switching from on to off, and to demonstrate this works in tandem with the variation of the soil moisture (resistance), I simulated both in the same plot:

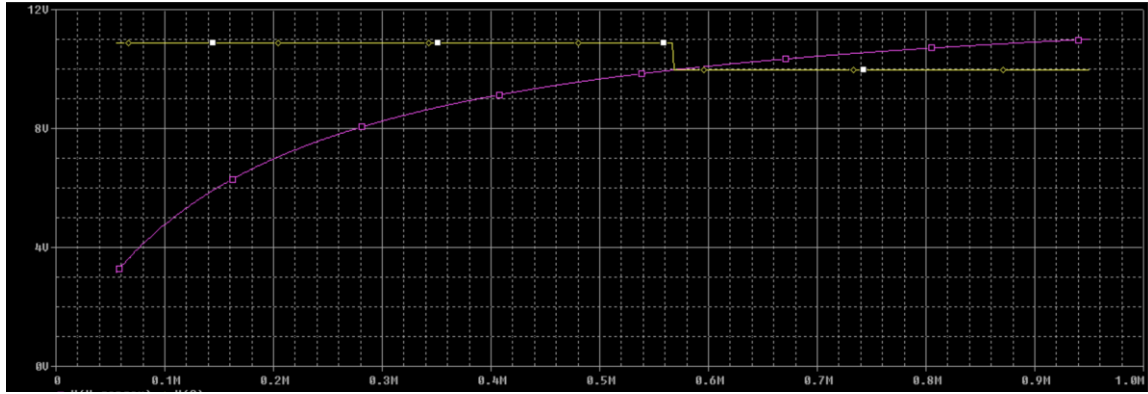


Figure 9. Proof of the window.

Trace Color	Trace Name	Y1
	X Values	568.376K
CURSOR 1,2	V(V_sensor)	9.971
	V(2)	9.963

This demonstrates that the resistors are correctly computed before entering the feedback loop.

Applying Millman's theorem to the non-inverting input at the moment of switching we get this relation:

$$V^+ = V_{ref}$$

$$V_{thH} = V_{ref} \times \left(1 + \frac{R_{in}}{R_{feedback}}\right) \quad (8)$$

$$V_{thL} = V_{ref} \times \left(1 + \frac{R_{in}}{R_{feedback}}\right) - V_{OH} \times \left(\frac{R_{in}}{R_{feedback}}\right) \quad (9)$$

Now we deduce R_{in} and $R_{feedback}$:

$$\begin{aligned} \Delta V_{hys} &= V_{OH} \times \frac{R_{in}}{R_{feedback}} \rightarrow 1.035 = 11.5 \times \left(\frac{R_{in}}{R_{feedback}}\right) \rightarrow \\ &\rightarrow \frac{R_{feedback}}{R_{in}} = \frac{11.5}{1.035} = 11.11 \end{aligned}$$

We choose a standard value for one of the unknowns. In my case I chose, $R_{in} = 10k\Omega$ and we find out the other unknown.

$$R_{feedback} = R_5$$

$$R_{feedback} = 11.11 \times 10k\Omega = 111.1k\Omega$$

Consequently, we look for V_{ref} which will be applied to the inverting input to center the hysteresis window. For this we use formula (8):

$$11.033 = V_{ref} \times \left(1 + \frac{10}{111.11}\right) \rightarrow$$

$$11.033 = V_{ref} \times 1.090 \rightarrow V_{ref} = 10.12V$$

The reference voltage is generated from the V_{cc} by using a voltage divider as discussed in <https://howtomechatronics.com/how-it-works/electrical-engineering/schmitt-trigger/>.

$$V_{ref} = V_{cc} \times \frac{R_1}{R_1 + R_3} \rightarrow 10.12V = 13V \times \frac{R_1}{R_1 + R_3} \rightarrow$$

Because we have 2 unknowns, we choose again a standard value. Therefore, $R_1 = 10k\Omega$.

$$\rightarrow 10.12 \times (10k\Omega + R_3) = 13 \times R_3 \rightarrow 101.2 + 10.2R_3 = 13R_3 \rightarrow 2.88R_3 = 101.2 \rightarrow$$

$$\rightarrow R_3 = 35.13k\Omega$$

Even though the calculations are mathematically correct, when implementing it into the circuit and carrying out the needed simulations, the hysteresis does not look as we would like it to appear, like a perfect square that switches instantly when $R_{sensor} = 566k\Omega$. Instead, it looks like at the switching resistance, it goes exponentially upwards. This happens due to the voltage drop of the LM358.

By adjusting the resistor values across multiple simulation tries, the components were increased in value until the comparator achieved the precise square-wave response required by the project specifications. As to why we increase the value is due to the losses the LM358 causes, so we increment the values to have a margin error.

Resistor	Theoretical Value	Adjusted Value
R_1	$10k\Omega$	$13.3k\Omega$
R_3	$35.13k\Omega$	$64.2k\Omega$
R_5	$111.1k\Omega$	$136k\Omega$

I ran the same DC sweep used in figure (3) to get the waveform below.

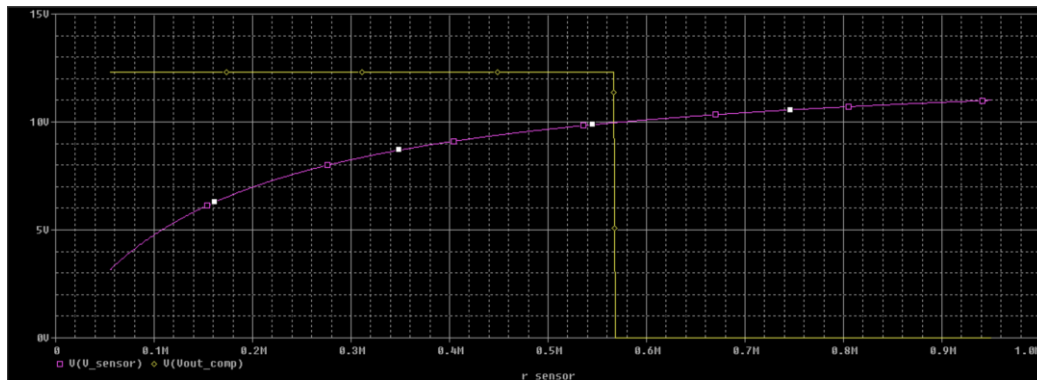


Figure 9. Signal at the output of the comparator.

We can perceive that the switching happens exactly at 30% ($566k\Omega$).

On the other hand, it is of utmost importance to demonstrate that the hysteresis happens. For this, I appended 2 files of different simulations.

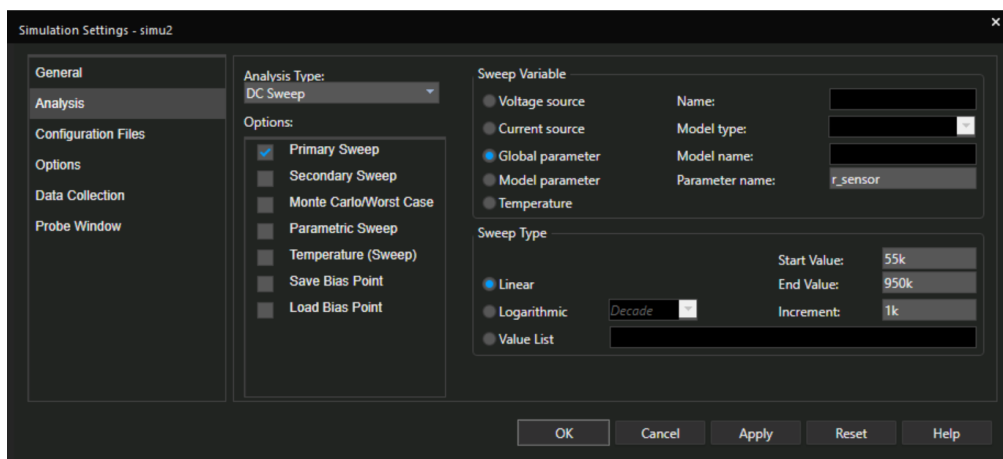


Figure 10. DC Sweep 2 for testing hysteresis.

Considering DC Sweep 1 from figure (3) and DC Sweep 2, we appended the simulations to see how the comparator compares the voltages and how it switches. This demonstrates that the pump repeats the cycle when detected humidity or dryness in the soil.

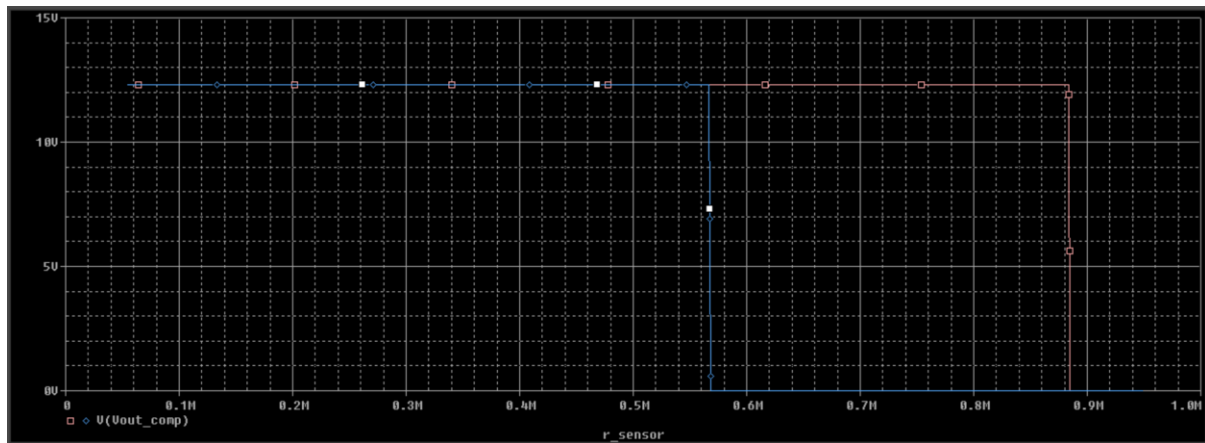


Figure 11. Demonstration of functionality. Hysteresis.

We can perceive that at $566k\Omega$ on the X-axis, our pump turns off, and at $950k\Omega$ (almost the twice the value) turns on. This demonstrates that the pump follows a cycle, and not only has 2 states, but knows how to keep switching and when. It is worth mentioning that the switching does not occur exactly at the value we want, yet this is due to non-idealities.

4.3 Buffer.

4.3.1 Circuit snippet of the stage.

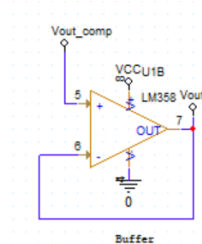


Figure 12. Buffer Stage.

4.3.2 Theory approach.

By ensuring that the mathematical calculations remain intact, I integrated a buffer cascaded between the Schmitt Trigger output and the electromechanical relay load. In this configuration, the output voltage directly follows the input voltage, therefore the gain is 1. According to microelectronic device theory by Rashid⁴, driving a low-impedance load directly from the comparator forces the operational amplifier into current saturation, causing a loss of the output voltage. Therefore, the buffer stage provides an isolation via a high impedance to the feedback network to preserve the calculated thresholds, while effectively sourcing the necessary current to the relay coil.

This is ideal when the goal is to transfer a signal from a high-impedance source to a low-impedance load without altering the signal's amplitude, just like in our case. On the other hand, as the output signal has very low impedance, it can be used directly to drive LEDs, which will aid us later.

4.3.3 Simulation.

To prove the signal is repeated at the output, I ran again the same DC sweep from figure (3).

⁴ M. H. Rashid, Ed., *Power Electronics Handbook*. San Diego, CA: Academic Press, 2001

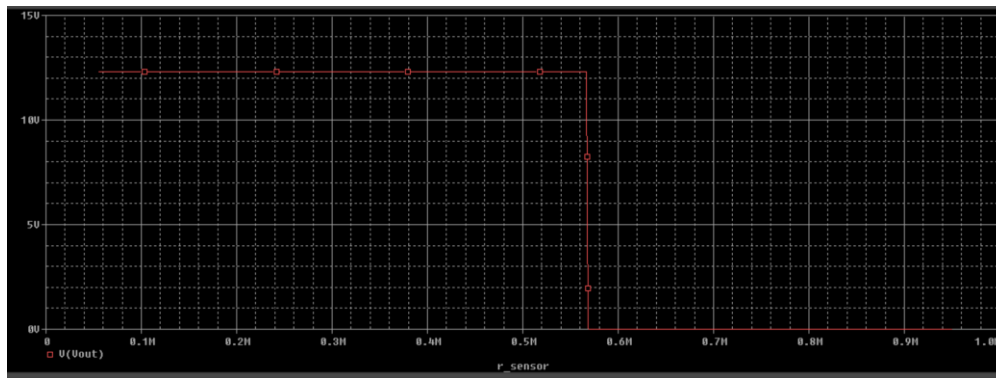


Figure 13. Output signal buffer.

4.4 Power electronics stage.

4.4.1 Power connections snippet (relay, LED, BJT).

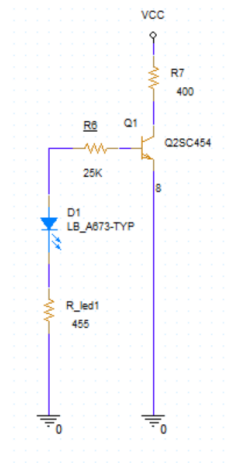


Figure 14. LED and relay network.

The LED and relay are connected in parallel so that the LED turns on simultaneously with the pump.

4.4.2 Theory approach, and calculation for LED.

In this part, I will discuss how to safely drive the LEDs from the buffer stage in order to work properly.

We will need a protection resistor with the finality of limiting the current which we can put before the LED or after hence as they are in series $I_{LED} = I_{resistor}$. The resistor is calculated using Ohm's law, therefore, by subtracting the LED's forward voltage drop V_{fwd} (around 3.9V for a blue LED with a current of 20mA based on datasheet) from the V_{cc} , we find the voltage that must be dropped across the resistor. Dividing this by the desired operating current, we obtain the value we need for the resistor:

$$R_{LED} = \frac{V_{cc} - V_{fwd}}{I} \quad (10)$$

Formula 10. Protection resistor calculation.

$$R_{LED} = \frac{13V - 3.9V}{0.02A} = 455\Omega$$

Consequently, I dove into the simulation of the current through the LED, and noticed the current is slightly less than $20mA$. This is given hence, even though we used the typical $3.9V$ value, in the datasheet it is specified it can go up to $4.5V$.

However, in order to check if the LED operates in that range, I ran some calculations:

$$V_{LED} = V_{cc} - (I_{measured} \times R_{LED}) \quad (11)$$

Formula 11. Ohm's Law for measured voltage.

$$V_{LED} = 13 - (0.0197 \text{ A} \times 455\Omega) = 4.0365V$$

This falls completely in the $3.9V$ - $4.5V$ range.

4.4.3 Simulations for LED.

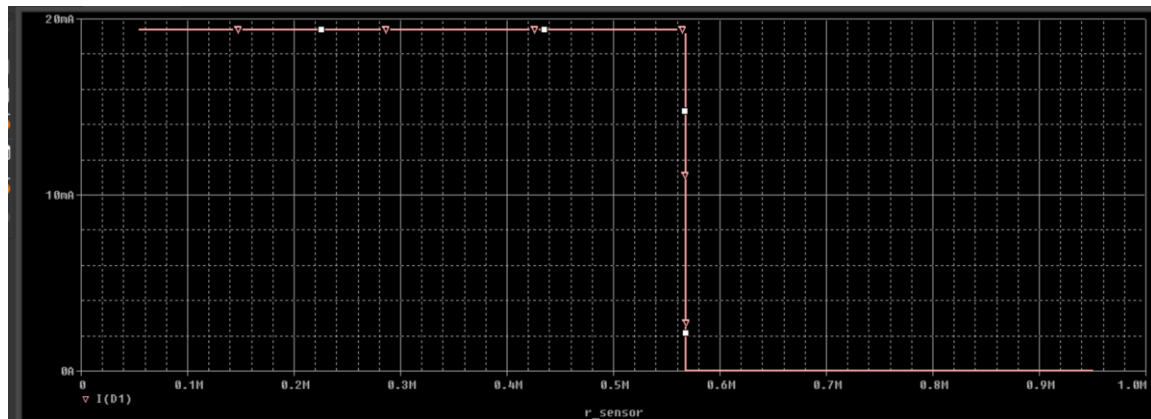


Figure 14. Current through the LED.

As detailed above, we have $\cong 19.7mA$, this still allows for high luminosity of the light. As a small remark, it is better for a LED to operate slightly less from its maximum value with the finality of keeping the components from getting warm or burning out.

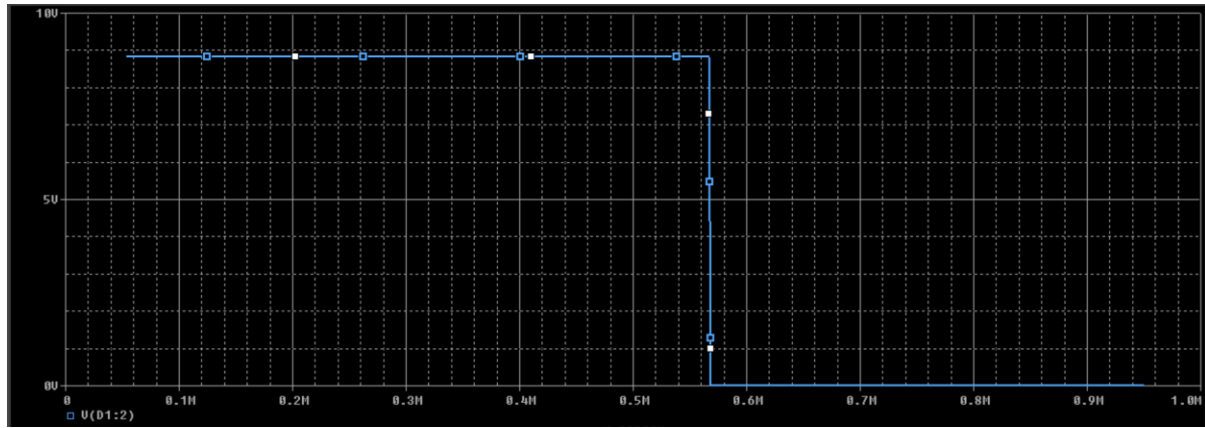


Figure 15. Output voltage LED.

From my $V_{cc} = 13V$ we subtract the voltage the LED consumes:

$$V_{cathode} = 13V - 4.0365V = 8.9635V \quad (12)$$

This is exactly what we can see in Figure 15.

As we can perceive the LED functions perfectly in tandem with the comparator, maintaining its range.

4.4.4 Theory approach, and calculation for relay.

A relay coil requires significantly more current to generate the magnetic field needed to close the switch. Therefore, if connected directly, the relay would draw too much current and permanently damage the circuit.

A BJT, like the 2SC454 I used, solves this by acting as an electrical switch. I choose the 2SC454 because, based on the datasheet, it possesses a maximum for the collector to emitter breakdown voltage of 30V, and an absolute maximum current of 100 mA. This provides a safety margin from the 20mA for the LED as well as for the 40mA for the relay.

Moreover, BJTs are widely utilized in power electronics as controllable switches to safely manage current flow between a source and a heavier load⁵. By feeding a very small current from buffer into the transistor base, it allows a much larger current to flow directly from the

⁵ Jiya, Immanuel & Gouws, Rupert. (2020). Overview of Power Electronic Switches: A Summary of the Past, State-of-the-Art and Illumination of the Future. Micromachines.

13V power supply, through the relay coil, into the collector, and down to ground, obviously through the resistor.

As heavily emphasized by Rashid, utilizing a BJT effectively as a switch requires intentionally driving it out of its active amplifying phase and into the saturation region. In saturation, the transistor is fully on. This minimizes the internal voltage drop across the transistor, ensuring maximum power goes to the relay.

From a datasheet for a relay that works at $V_{cc} = 13V$ the value of the relay coil is 400Ω .

Therefore, my collector current should be:

$$I = \frac{V}{R} = \frac{13V}{400\Omega} = 32.5mA$$

Now that we know the current that flows for the relay, we can calculate the resistance value for the base of the transistor. Considering that for my kind of transistor my gain $\beta=100$:

$$I_c = \beta \times I_B \rightarrow I_B = \frac{I_c}{\beta} = \frac{32.5mA}{100} = 325\mu A$$

$$R = \frac{13V - 4.04V}{325\mu A} \cong 28k\Omega$$

4.4.5 Simulation for relay.

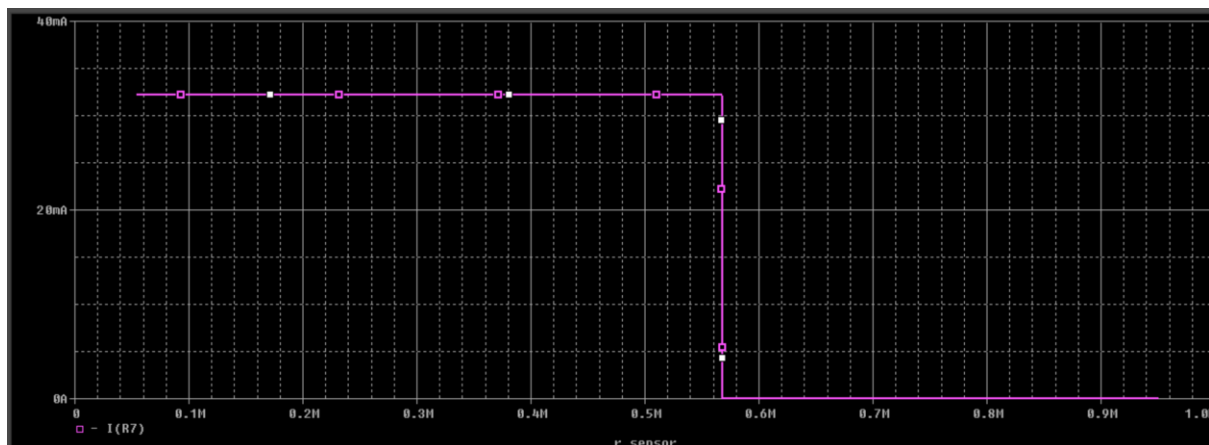


Figure 16. Current through relay.

5. Monte Carlo Statistical Analysis.

While mathematical calculations use fixed, ideal component values, discrete physical components have specified manufacturing tolerances which can take a component of 10k with 5% tolerance and make it vary between 2 relatively close values. The effect of these variations can cause the circuit to deviate from expected behavior. The Monte Carlo analysis is used to predict the overall exactness and reliability of the irrigation controller by simulating the circuit's response when all device parameters are randomly varied within their tolerance limits.

In Monte Carlo analysis, each component defined with a tolerance will have a different value in each simulation run. During the simulation, PSpice generates random values for the components with defined tolerance based on the "random number seed" parameter in the simulation, which represents the initial value used by the random number generator. The simulation is run multiple times, with each run using a different set of values for the components. On the X-axis is my resistance range (k Ω), and on the Y-axis is the output voltage(V(out)). Each colored curve represents the result of a Monte Carlo simulation, indicating how the circuit's performance varies due to resistance variations.

I started with a tolerance of 10% for every resistor in my circuit in order to have a starting point from which I could vary. Yet first this is how I defined my Monte Carlo simulation:\

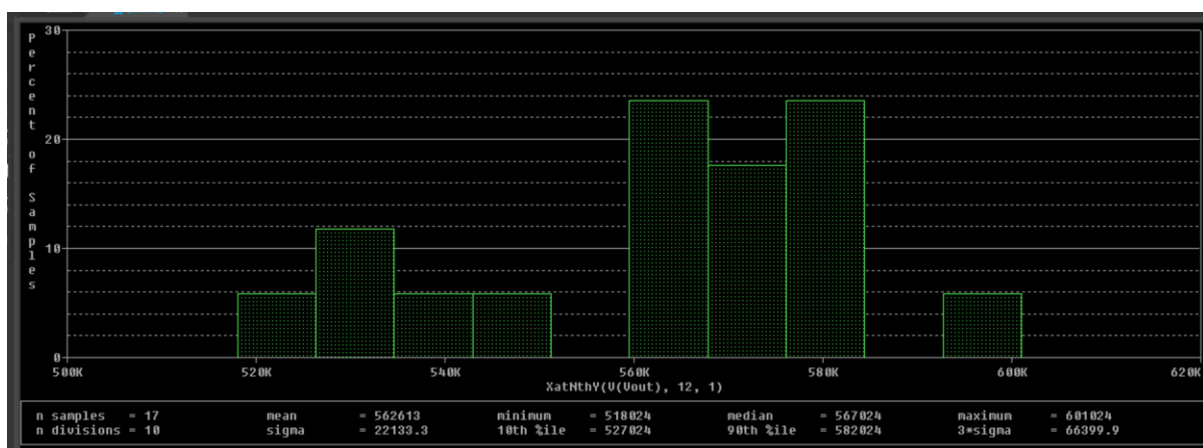


Figure 17. Monte Carlo 10% tolerance.

Because the range from 560k varies left and right with 40k, the probability for my resistors to not be stable and to distort the circuit's functionality is very likely. Therefore, I sized down to 5% for every resistor.

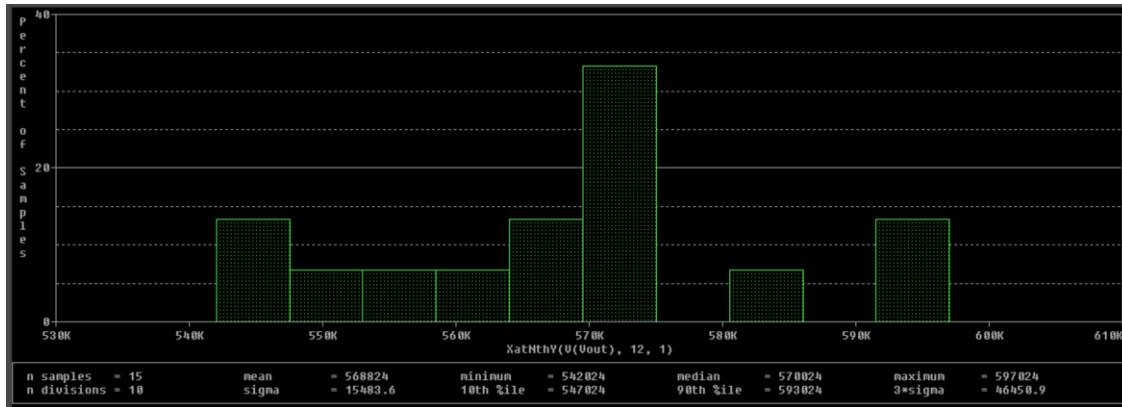


Figure 17. Monte Carlo 5% tolerance.

This simulation is not good either hence my variation is still large. However, I set the voltage divider resistance's tolerance of V_{ref} to 1%.

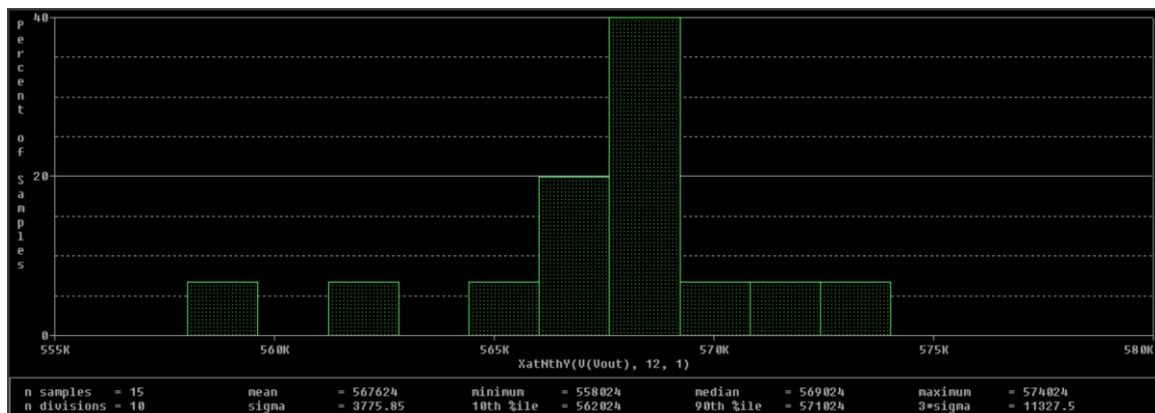


Figure 18. Monte Carlo 1% tolerance V_{ref} .

In the end, I decided to set my resistances for my comparator to 0.1% tolerance. I do this hence when running the Monte Carlo simulation I get the best results by doing that. However, for the rest of the resistors at the stage of power I left it at 10% tolerance because they do not affect the variation.

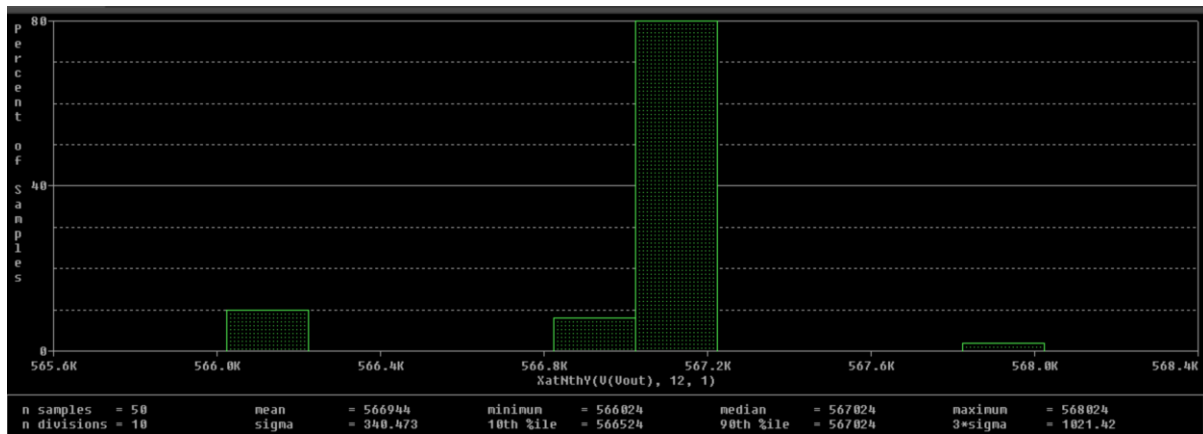


Figure 19. Final Monte Carlo Analysis.

As we can perceive, the range is close to the value I calculated for the switch. For the same reason, we know it works like that hence my probability for the resistor to work at the value I calculated is greater than the ones in the vicinity.

6. Worst Case Analysis.

I performed a Worst Case to find the maximum deviation at the comparator's output. The analysis ran the individual's sensitivity of each component and subsequently forced all components to their opposing tolerance limits. The resulting summary indicates a worst-case shift of 12.27 Volts occurring at a sensor resistance of 568k Ω .



Figure 20. Worst Case scenario output file.

I analyzed the performance in the less favorable conditions, taking into account when the switch takes place, and considering the tolerances applied.

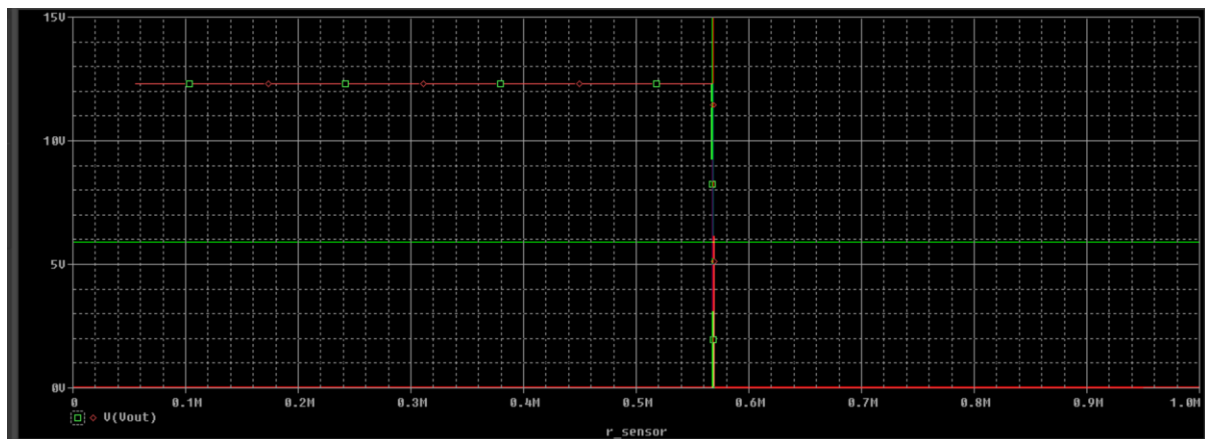


Figure 21. Simulation cursor on deviation.

Trace Name	Y1	Y2	Y1 - Y2
X Values	568.376K	567.521K	854.670

Figure 22. Measurement between the values.

Lastly, I applied two cursors to measure the values between the variations of the resistances, in this case, it is a variation of 854.670Ω .

7. Design Enhancements.

7.1 Initial circuit.

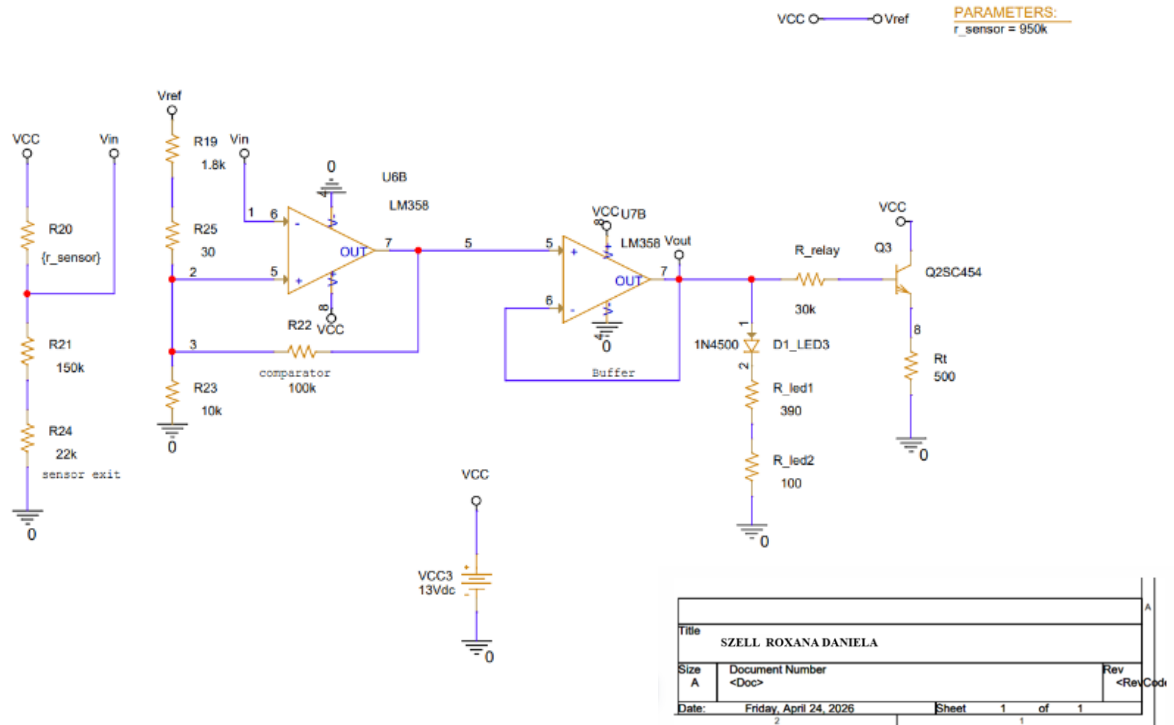


Figure 23. Initial circuit.

7.1.1 Modifications.

After countless tries, which will be made known at the presentation, I reached this semifinal circuit. However, it was not as good as it could be.

Furthermore, I undergone some changes based on some feedback, and further examination until I reached my final circuit.

Conversion Stage.

First and foremost, the most important thing I overlooked was the conversion stage which was not implemented in my semifinal model. As we can perceive, the voltage divider goes directly into the inverting pin of the comparator. This can produce the drag of the current by the comparator and carry out loading effects because of the high resistance values we are using, causing inaccurate switching.

The finality we desire is to have a stable signal before going into the comparator. Therefore, the solution I implemented was adding a high impedance buffer stage (which prevents the loading effect) configured as a voltage follower, which repeats the voltage from the voltage divider, preserving the exact mathematical value I calculated thanks to the low output impedance of the op amp.

Resistance values and tolerances.

Another factor I altered was the resistance values for the second part of my circuit. Initially the switching was done too early, so I manually adjusted each resistance for the feedback of the comparator.

Load transistor.

After measuring the current through R7 I realized the load cannot be between the emitter and ground because I got a significantly lower current than calculated (half of 32.5mA). Placing the load on the emitter prevents the transistor from fully saturating, leaving it stuck in the active region and stops acting like a switch.

I solved this issue by trying out a different connection after looking into an article⁶ on BJT configurations. I realized my error when I read: "common Emitter or grounded emitter configuration". That is when I realized that the emitter goes connected to the GND, and the load before the collector.

7.2 Final circuit.

⁶ Electronics Tutorials. (2019, February 4). Bipolar Transistor Tutorial, The BJT Transistor. Basic Electronics Tutorials. https://www.electronics-tutorials.ws/transistor/tran_1.html

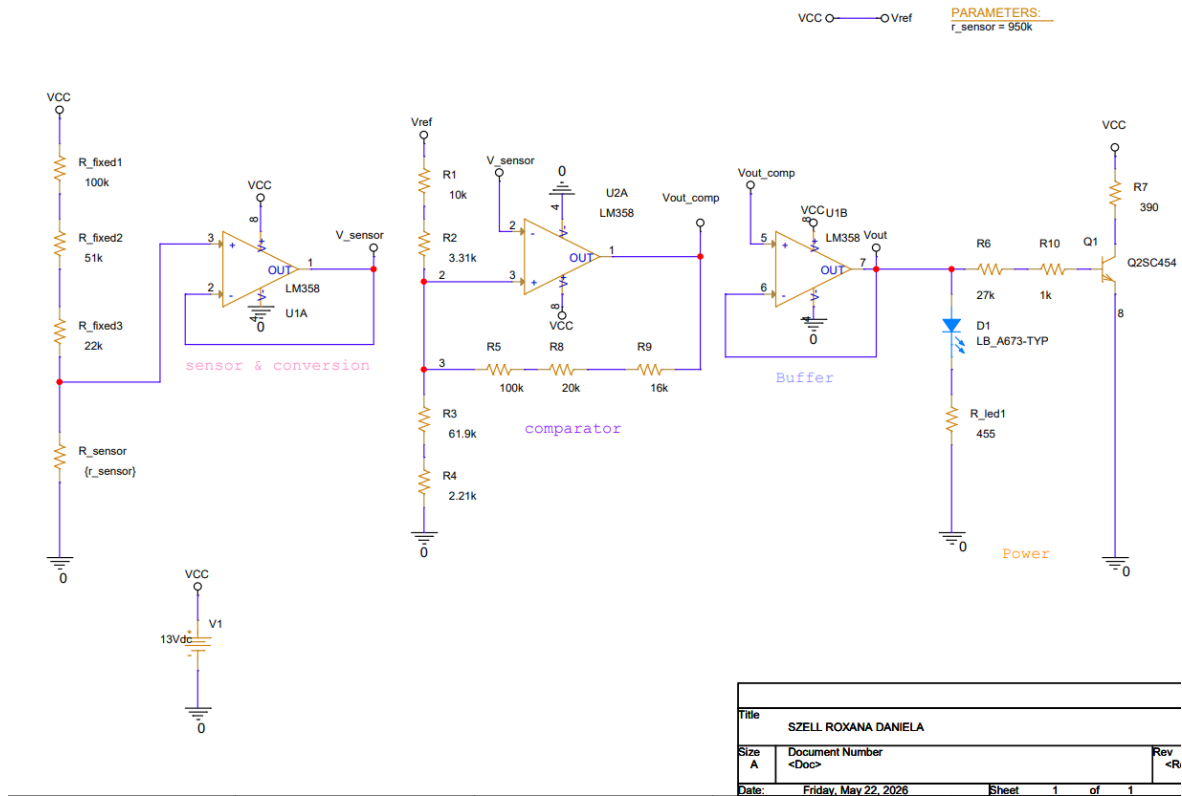


Figure 24. Final circuit.

8. Bill of Materials.

Quantity	Component	Value	Tolerance	Datasheet	Price
3	OP-AMP	LM358	-	LM358 DATASHEET	1,57 RON/1
1	R_fixed1	100k Ω	-	res DATASHEET	0,05 RON
1	R_fixed2	51k Ω	-	res DATASHEET	0,05 RON
1	R_fixed3	22k Ω	-	res DATASHEET	0,05 RON
1	R_sensor	FC-28	-	SENSOR DATASHEET	6,5 RON
1	R1	10k Ω	0.1%	0.1% res DATASHEET	2,10 RON
1	R2	3.31k Ω	0.1%	0.1% res DATASHEET	2,10 RON
1	R3	61.9k Ω	0.1%	0.1% res DATASHEET	2,10 RON
1	R4	2.21k Ω	0.1%	0.1% res DATASHEET	2,10 RON
1	R5	100k Ω	0.1%	0.1% res DATASHEET	2,10 RON
1	R6	27k Ω	10%	10% res DATASHEET	0,26 RON
1	R7	390 Ω	10%	10% res DATASHEET	0,26 RON
1	R8	20k Ω	0.1%	0.1% res DATASHEET	2,10 RON
1	R9	16k Ω	0.1%	0.1% res DATASHEET	2,10 RON
1	R_led1	455 Ω	10%	10% res DATASHEET	0,26 RON
1	Q1	Q2SC454	-	Q1 DATASHEET	0,79 RON
1	LED	D1 LB_A673- TYP	-	LED DATASHEET	2,62 RON
1	Battery	V1	-	BATTERY DATASHEET	78,70 RON

TOTAL COST: 108,95 RON.

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10. Conclusion.

In conclusion, to bring this project to an end, I demonstrated the design and implementation of an automated irrigation system efficiently, and taking into account paramount factors such as energy, power, and simplicity.

The system successfully processes real sensor data to determine watering needs for our soil, effectively mitigating the risks of both overwatering and underwatering. I do this by activating the pump only when specific environmental thresholds are met, the project provides a practical solution for water handling activities.